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Metagenomic Analysis of Fecal Samples as a Screening Tool for Colorectal Cancer

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Abstract

Motivation: Colorectal cancer (CRC) is a leading cause of cancer-related deaths worldwide, with high survival rates if detected early. Despite advances in screening methods, a significant portion of the population prefers non-invasive options over colonoscopy due to its invasiveness. This study explores metagenomic analysis of fecal samples as a non-invasive alternative for CRC screening by examining changes in the gut microbiome associated with CRC.

Results: Using fecal metagenomic data, convolutional neural network (CNN) and multilayer perceptron (MLP) models were developed to classify samples into Healthy, Adenoma, and Carcinoma groups. CNN achieved the highest performance, with a precision of 0.97, recall of 0.98, and F1-score of 0.98, outperforming baseline Random Forest and Support Vector Machine classifiers. Feature extraction utilizing saliency maps revealed key bacterial species influencing predictions, including *Veillonella parvula* and *Akkermansia muciniphila,* which are associated with factors influencing CRC. These findings demonstrate the potential of neural networks to improve the accuracy of non-invasive CRC screening.

Availability: All data sets are publicly available in full at https://www.ebi.ac.uk/ena/browser/view/PRJEB6070 and https://tcma.pratt.duke.edu/ and all scripts can be found on github at https://github.com/bbock27/Comp_genomics_final

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1 Introduction

World wide, colorectal cancer is the third most commonly diagnosed malignancy and is the second most common cause of cancer deaths annually [1]. The standard development of colorectal cancer is a stepwise process where localized precancerous adenomatous polyps (known as adenomas) develop in the colon and progress into invasive and cancerous tumors (known as carcinomas) [2]. The development of carcinomas can largely be prevented if adenomas are detected and removed, with the survival rate of colorectal cancer exceeding 90% if the diagnosis and treatment occur while it is still localized [3].

Death rates from colorectal cancer have declined steadily in the past 40 years, mainly due to advances in highly sensitive screening methods such as fecal occult blood testing (FOBT), sigmoidoscopy, and colonoscopy. In particular, colonoscopies are a particularly potent tool against colorectal cancer as they allow for full examination of the bowel with the opportunity for same-session colonic biopsies and removal of polyps. However, surveys indicate that 50-60% of adults prefer non-invasive screening methods [4]. This may be due to the large inconvenience and invasiveness of colonoscopy procedures. With colorectal cancer treatment benefitting from early detection, effective non-invasive screening methods are in high demand. One of the most common methods of non-invasive screening is the guaiac fecal occult blood test (gFOBT) which detects blood in a fecal sample, indicating a possible advanced adenoma and carcinoma. gFOBT testing is a particularly well-suited tool for mass screening as it has high sensitivity and is very cheap and convenient for the patient. In particular, the cost and simple process contribute heavily to the likelihood of at-risk populations participating in regular screenings. However, this test has a high false positive rate as a wide variety of other disorders can cause blood to be present in feces [5]. The standard procedure once a screening test has returned a positive result is to follow up with additional tests, usually involving at least one more round of non-invasive tests. The final definitive diagnosis is generally made after a colonoscopy is performed. More accurate screening tests would reduce the medical resources used on false positive cases and decrease the number of people who have to undergo invasive procedures following false positive results.

The human microbiome represents the entire microbial population that colonizes the human body. The gut microbiota in particular has a vital function in maintaining overall health and has been related to the emergence of conditions such as type 2 diabetes, colorectal cancer, and obesity. Several findings focused on the complication and two-way correlation between microbiota and cancer. Cancer progression may alter the microbiota. Meanwhile, changing the microbiota may have an impact on cancer development. Many studies have confirmed that susceptibility to CRC or tumor progression is affected by changes in the gut microbiome through mechanisms such as influencing inflammation levels, DNA damage, and harmful microbial metabolite production [6]. These studies suggest a strong link between the gut microbiota and cancer progression, and a possibility to use changes in the gut microbiota as a screening tool for colorectal cancer.

The advances in high throughput sequencing methods have made this option a possibility. High throughput sequencing generates large amounts of data that require powerful computational tools to provide accurate analysis of data. This makes machine learning a major approach to data analysis related to the cancer-associated microbiome. Recent examples of these methods have applied machine learning algorithms to the classification of cancers. A random forest model was implemented for classification of colorectal cancer based on microbial information from 490 patients obtained by 16S rRNA sequencing [7]. With a reported detection rate of 91.7% of cases, this method exceeded existing screening methods and shows the potential of machine learning models in this application.

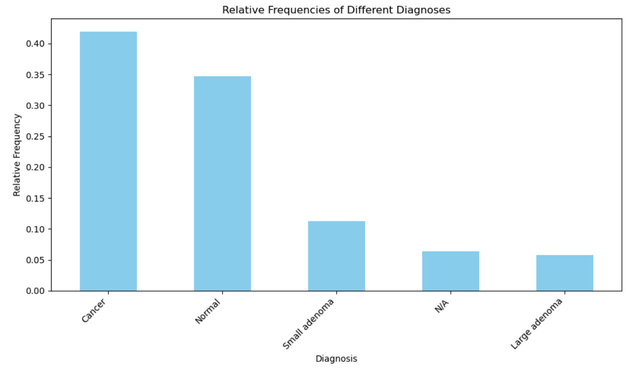
However, the space of applying neural networks to this problem is underdeveloped and in its infancy. In this work we used fecal sample Illumina paired-end sequencing reads obtained through the European Nucleotide Archive (ENA) and The Cancer Microbiome Atlas (TCMA) to develop a convolutional neural network (CNN) model and a multilayer perceptron (MLP) model that can predict the disease state in regards to colorectal cancer of a patient from fecal samples, advancing the potential for non-invasive, highly accurate diagnostic tools.

2 Methods

2.1 Data description

The data acquired through TCMA can be found at https://tcma.pratt.duke.edu/ and the data from the ENA can be found under the accession number ERP005534. The data from TCMA was obtained in the form of relative taxonomic profiles and a total of 53 samples were obtained. The data acquired from the ENA was in the form of raw Illumina paired-end reads. This data set had a total of 424 samples for a total of 477 samples between the two datasets. The metadata for each sample from the ENA includes disease state, age, sex, BMI, and many other fields and can be used for additional benchmarking of the models.

Fig. 1. Relative abundances of classes of training data.



2.2 Pre-processing

The data from the ENA was in the form of FASTQ files and taxonomic profiles had to be created from them. This was accomplished by utilizing the metagenomic classifier Kraken2. The metadata between the datasets was congregated and the relative abundance of each disease state was calculated and plotted (Fig. 1). 40% of the samples were from individuals with colorectal cancer, 34% of the samples were from healthy individuals and 16% of the samples were from individuals with varying sizes of adenomas. The remaining samples had an unlabeled disease state and had to be discarded from the data set.

The quality of the sequencing reads was checked using FastQC (v 0.11.7) [8] and MultiQC (v 1.9) [9]. Then, BBduk (v 38.26) [10] was used to clean the reads, removing sequencing adapters, low-quality ends of reads (PHRED score < 20), and short reads after trimming (pairs where one of the reads was length < 50) were removed.

The clean reads were classified utilizing Kraken2 (v 2.1.3) [11], a computational tool designed for the high-throughput taxonomic classification of sequences. Kraken2 employs a k-mer–based approach to assign taxonomic labels to reads by comparing them against a reference database of known genomes. We used the standard kraken2 database built from NCBI taxonomic information and the complete genomes in RefSeq for the bacterial, archaeal, and viral domains, along with the human genome and a collection of known vectors(UniVec_Core). A confidence threshold of 0.1 was used to reduce false-positive classifications and Kraken2 was run in paired-end mode to leverage read pairs for improved accuracy. Reads were processed in parallel to optimize runtime performance, utilizing the tool's multi-threading capability with 48 threads. This was followed by Bayesian re-assignment at the species level using Bracken [12], with the read length parameter set at 100.

Due to the nature of which Kraken2 and Bracken classify reads and estimate abundances, there are inevitably rare misclassified reads that can skew downstream analysis and must be filtered out. This can be done by removing low-count assignments and features with support in a small fraction of the total samples. After filtering, this resulted in a feature set of 344 total species.

Because of the small sample set of the large adenoma and small adenoma classes, they were joined to form the adenoma class. The samples were then partitioned into a testing and training dataset with a split of 67% training and 33% testing. The training dataset was then batched into a torch DataLoader (batch_size=32) with oversampling of the Normal and Adenoma classes in the training data in order to mediate the class imbalance.

2.3 Model Architecture

The first models trained were fandom forest and support vector machine models obtained through the scikit-learn python package. The random forest model was trained with 50, 100, and 200 estimators and evaluated on precision, recall, and F1 score. The support vector machine was trained with the linear, polynomial, and radial basis function kernels and their individual performance was evaluated based on the same metrics as the random forest model.

We employed two distinct neural network architectures that were created and evaluated using the torch python package. Cross entropy loss was utilized as the loss function and the adam function with a learning rate of .001 was used as the optimizer.

The first model is a shallow MLP that consists of 5 layers and uses the reLU activation function. The first layer is the input layer which is made up of a fully connected linear layer with 344 inputs and 64 outputs with an accompanying batch normalization layer with 64 features. Following this, a dropout layer (p=0.2) is applied to mitigate overfitting of the model. The fourth layer is a fully connected linear layer that reduces the dimensionality from 64 to 32. The last fully connected linear layer reduces the output into a probability of three classes: Cancer, Normal, and Adenoma. The MLP was trained for 60 epochs and the ideal generation was chosen to be 18 (fig. 2a).

The CNN model was designed to extract hierarchical features from one-dimensional input data and consisted of three convolutional-pooling blocks followed by fully connected layers for classification.

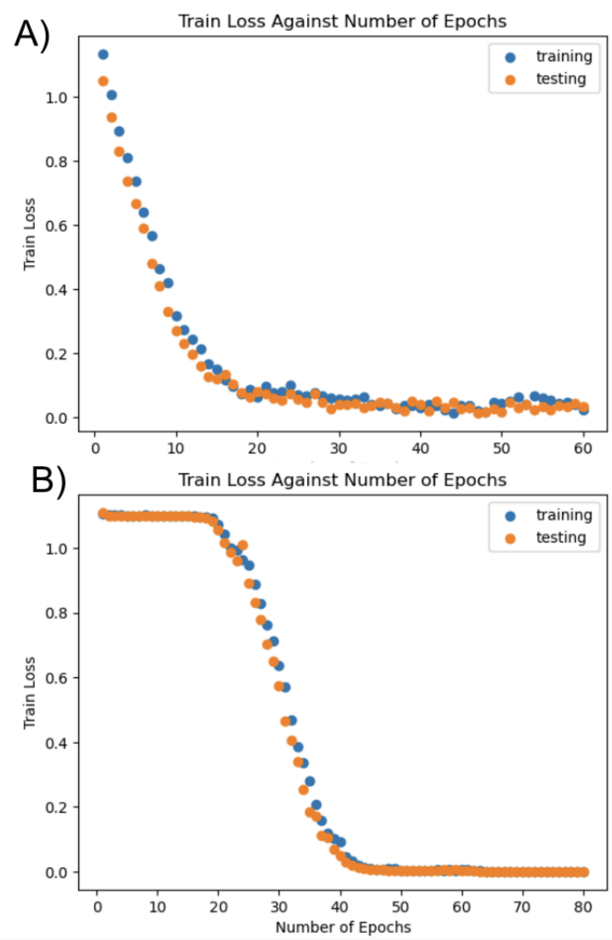
The first convolutional layer applied 32 filters with a kernel size of 4, a stride of 1, and no padding, followed by a max-pooling layer with a kernel size of 2 and a stride of 2. The second convolutional layer increased the filter count to 64, with the same kernel size, stride, and pooling configuration. The third convolutional layer further expanded to 128 filters, followed by another max-pooling operation. These convolutional layers progressively reduced the spatial dimensions of the input while extracting increasingly abstract features.

The output from the final pooling layer was flattened into a one-dimensional vector and passed through two fully connected layers. The first dense layer consisted of 128 units, followed by a ReLU activation function, while the second dense layer included 32 units, also followed by ReLU activation. The final output layer employed a linear transformation to map the feature representation to three output nodes, corresponding to the predicted class probabilities. The CNN was trained for 80 epochs and the ideal generation was chosen to be 40 (fig. 2b).

2.4 Feature extraction

To perform feature extraction of the CNN, saliency maps were calculated through backpropagation for each sample in the training set. The mean saliency for each feature within a class was taken and outliers were chosen based on z-score with a threshold of 3.

Fig. 2. Relative abundances of classes of training data. a) the cross entropy loss of the MLP per epoch. b) the cross entropy loss of the CNN per epoch



3 Results

3.1 Model Evaluations

In order to evaluate the effectiveness of the models, we looked at the precision, recall, and f1-score of all four of the models (fig 3). As we can see from the table, the MLP and CNN performed significantly better than the random forest and support vector machine, with the CNN performing slightly better than the MLP.

Fig. 3. precision, recall, and f1-score of the different models. a) The random forest classifier b) the support vector machine c) the MLP d) the CNN

A)

	precision	recall	f1-score
accuracy			0.51
macro avg	0.17	0.33	0.22
weighted avg	0.26	0.51	0.34

B)

	precision	recall	f1-score
accuracy			0.48
macro avg	0.36	0.33	0.28
weighted avg	0.39	0.48	0.37

C)

	precision	recall	f1-score
accuracy			.95
macro avg	0.97	0.94	0.96
weighted avg	0.96	0.95	0.95

D)

	precision	recall	f1-score
accuracy			.98
macro avg	0.97	0.97	0.97
weighted avg	0.99	0.98	0.98

The support vector machine and random forest models had very poor macro and weighted average scores consistent with predicting the same state for each sample indicating these models were not able to elucidate the relationship in the data.

3.2 Feature Extraction

To identify the most influential features contributing to model predictions, we utilized saliency maps generated from the CNN. These maps provide insight into which microbial features were most significant in differentiating between disease states. For each sample in the training set, we computed the saliency map using backpropagation to trace the impact of each feature on the model's decision(fig 4). Then we looked at which features contributed most heavily to predicting each class. The saliency maps revealed that the features corresponding to specific bacterial taxa, such as *Anaerostipes hadrus* and *Faecalibacterium prausnitzii*, were highlighted as most influential(fig 5). These taxa have been previously linked to colorectal cancer and increased inflammation [13] [14], suggesting their critical role in the model's ability to distinguish between cancerous and non-cancerous states has a biological basis. A total of six species were identified to be most influential for a diagnosis of cancer (z-score > 3). For a healthy diagnosis, four species were identified to be most influential and seven were found for an adenoma diagnosis.

Fig. 4. Saliency of CNN visualization. Each row represents a different sample and each column represents a different species of bacteria. The coloring represents how much that species contributed to the classification result. The darker the color is, the greater an influence that feature is on the decision. A near-solid red vertical line indicates a feature that is highly influential across all samples.

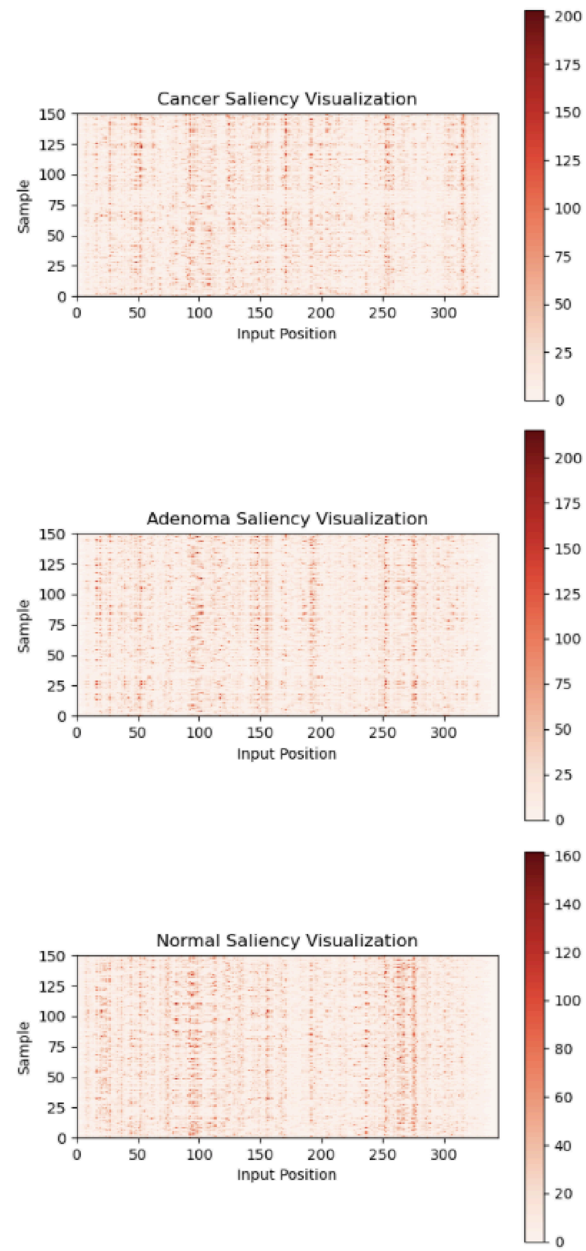


Fig. 5. Relative abundances of classes of training data. a) the cross entropy loss of the MLP per epoch. b) the cross entropy loss of the CNN per epoch

Disease State	Correlated Bacteria
Carcinoma	*Veillonella parvula* *Faecalibacterium prausnitzii* *Anaerostipes hadrus* *Parabacteroides distasonis* *Alistipes shahii* *Blautia massiliensis*
Healthy	*Akkermansia muciniphila* *Collinsella aerofaciens* *Sutterella wadsworthensis* *Haemophilus parainfluenzae*
Adenoma	*Veillonella parvula* *Anaerostipes hadrus* *Akkermansia muciniphila* *Bacteroides xylanisolvens* *Blautia coccoides* *Dialister pneumosintes* *Sutterella wadsworthensis*

4 Discussion

4.1 Model Performances

The superior performance of the MLP and CNN models can be attributed to their ability to handle complex, high-dimensional data effectively. Unlike traditional models such as Random Forest and Support Vector Machine which may struggle with the intricacies of microbiome data, MLPs and CNNs are adept at capturing and leveraging hierarchical patterns within the data. CNNs are particularly well-suited for this task due to their convolutional layers, which automatically learn and extract spatial features from microbial profiles, capturing patterns such as relative abundance, rare taxa presence, and interrelationships among different microbial communities. MLPs, on the other hand, can learn complex, non-linear relationships through multiple layers of dense connections, enabling them to better represent the underlying structure of the data. These models can adapt to the non-linearities and high-dimensional nature of metagenomic data without requiring manual feature engineering, a common limitation of traditional methods. The training process of MLPs and CNNs involves stochastic gradient descent and backpropagation, which allows them to adjust weights iteratively and find optimal solutions, leading to better generalization to unseen data. This flexibility and capability for automated feature learning make MLP and CNN models particularly effective for classifying CRC from fecal samples, outperforming Random Forest and SVM models which rely on predefined feature extraction and are less suited for handling complex, non-linear data structures.

4.2 Extracted Features

Feature extraction from the CNN revealed key insights into the microbiome profiles associated with different disease states. Saliency maps generated through backpropagation identified microbial taxa that were most influential in model predictions for each class (Carcinoma, Adenoma, and Healthy). These findings are consistent with prior studies linking specific bacterial species to colorectal cancer (CRC) progression, adenoma formation, and healthy gut microbiota.

For the Carcinoma class, six species were identified as having a significant influence on predictions: *Faecalibacterium prausnitzii, Anaerostipes hadrus, Parabacteroides distasonis, Alistipes shahii, Veillonella parvula,* and *Blautia massiliensis*. Several of these taxa, such as *Faecalibacterium prausnitzii* and *Anaerostipes hadrus*, have been implicated in inflammation and CRC [13] [14]. Notably, *Faecalibacterium prausnitzii* has been linked to anti-inflammatory properties under healthy conditions but appears reduced in abundance in CRC [14], potentially disrupting gut homeostasis. *Veillonella parvula* is a bacteria that is known to mediate activation of the Nod2/CCN4/NF-κB signaling pathway [15]. Variation in activation levels of this pathway is associated with an increased risk of colon cancer, however the biological mechanisms behind it are not fully understood. Additionally, *Veillonella parvula* is a strictly anaerobic lactate-fermenting bacterium. High tumor lactate levels directly associate with high tumor growth, metastasis, and patients' poor prognosis[16], indicating *Veillonella parvula* has the potential to be a strong indicator of a tumor's stage of development and severity.

The Healthy class was most strongly influenced by *Akkermansia muciniphila*, *Collinsella aerofaciens*, *Sutterella wadsworthensis*, and *Haemophilus parainfluenzae*. These taxa are commonly associated with a balanced and functional gut microbiota. *Akkermansia muciniphila*, for example, is known for its role in mucin degradation and gut barrier maintenance, which may provide protection against CRC[17].

For the Adenoma class, seven species were identified as key predictors: *Anaerostipes hadrus*, *Akkermansia muciniphila*, *Bacteroides xylanisolvens*, *Blautia coccoides*, *Dialister pneumosintes*, *Veillonella parvula*, and *Sutterella wadsworthensis*. The overlap of certain taxa, such as *Anaerostipes hadrus* and *Veillonella parvula*, with both Carcinoma and Adenoma classes highlights their potential involvement in early CRC development. This overlap may suggest a progressive shift in microbial composition as adenomas transition into carcinomas.

The identification of influential taxa through saliency maps provides both biological and clinical relevance. These results underscore the potential of microbial biomarkers in non-invasive CRC screening. Specifically, species such as *Veillonella parvula* and *Akkermansia muciniphila* could serve as targets for diagnostic tools or therapeutic interventions. Furthermore, the ability of

CNNs to uncover these taxa without predefined feature selection highlights the advantages of deep learning in microbiome analysis.

4.3 Implications and Limitations

The findings demonstrate that changes in gut microbial composition are strongly associated with CRC and its precursors, making the microbiome a valuable resource for non-invasive diagnostic methods. However, one needs to be careful with the generalizability of these results as they are limited by the relatively small sample size and regional bias. The majority of the samples were sourced from individuals in northeastern Europe, in particular Germany and France. Due to the gut microbiome being highly variable across populations[18], it is likely that the model and subsequently the features extracted won't generalize well across a variety of populations. However, what can be generalized is the efficacy of neural networks in detecting CRC and the trends in what bacteria were identified as significant to predicting a cancerous state.

Future studies should aim to validate these findings across larger, more diverse cohorts and investigate the functional roles of bacteria involved in metabolizing materials associated with tumor growth such as lactic acid, and those involved in inflammation and DNA repair pathways as potential diagnostic and treatment options for CRC.

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